

GitHub Actions Basics

Run a workflow when source changes are pushed.

Learning Goals

- Distinguish workflows, triggers, jobs, and steps.
- Run a workflow and inspect its log.

Five Terms

Term	Meaning
Workflow	YAML instructions in <code>.github/workflows/</code>
Trigger	Event that starts the workflow, such as a push
Job	Work performed on a runner
Step	One command or reusable action in a job
Runner	A separate computer that runs the job

The runner needs your code and build tools.

Create Your First Workflow

The supplied file is at

`module3/code/3_Github_io/2_Github_io_Actions_Basics/github/workflows/hello.yml`.

- Its folder is named `github` ; copy it to `.github/` in **your own** repository.

```
name: My First Action
on: push
jobs:
  say-hello:
    runs-on: ubuntu-latest
    steps:
      - name: Greet the world
        run: echo "Hello from GitHub Actions!"
      - name: Show date
        run: date
```

Virtual Machine behind GitHub Actions

When a workflow runs, GitHub Actions starts a temporary Linux virtual machine (a runner) behind the scenes.

- The YAML file defines the steps this machine runs, such as installing tools, building the site, and deploying it.
- In this example, `runs-on: ubuntu-latest` selects an Ubuntu Linux machine, which runs the `echo` and `date` commands.
- The machine is discarded after the job finishes.

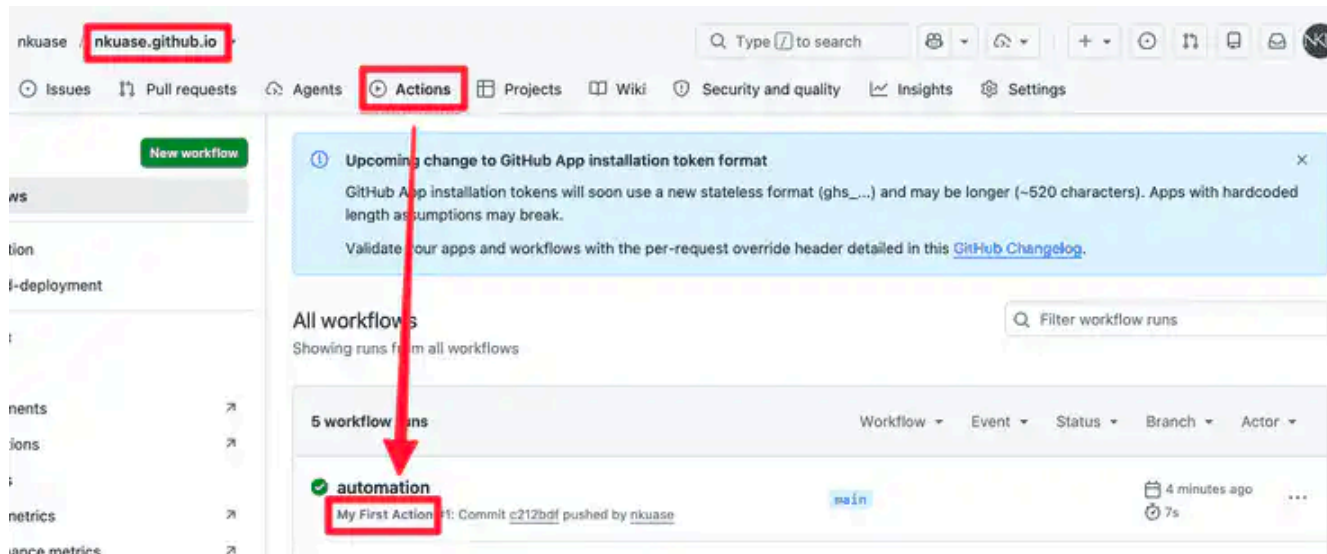
Commit, Push, and Inspect

You can commit and push through VS Code or command-line.

```
git add .github/workflows/hello.yml  
git commit -m "Add hello workflow"  
git push origin main
```

Open the **Actions** tab (next to Code and Pull requests) in your GitHub repository.

1. The **My First Action** is executed when we push to the github through github actions.
2. You can find the report from the **Actions** tab.



What This Workflow Does Not Do

The hello workflow prints text; it does not publish a website. For Hugo deployment, the workflow must:

1. Check out your source and install Hugo.
2. Configure Pages and build `public/`.
3. Upload the built files.
4. Deploy them to GitHub Pages.

Check Your Understanding

- Where must GitHub find a workflow file?
- What starts this workflow?
- Where can you find the first failed step?

Next: read the reusable actions in the supplied Hugo workflow.